

TOE (Dry, top-down): Unitary Interior + Gated Readout + Invariants + Explicit Tail Control

0. Non-claim (read once, then proceed)

This document is a *formal architecture*: a top-down operator framework with unconditional theorems that follow from stated axioms. It does *not* claim new proofs of the Riemann Hypothesis, $P \neq NP$, or Navier–Stokes regularity. Where those appear, we give *reformulations/reductions* inside the same architecture, clearly labeled.

I. Theory of Everything (TOE): one spine, zero mysticism

I.1. Primitive objects

Definition 1 (State space). A physical/model “interior” is represented by a complex Hilbert space $\mathcal{H}$.

Definition 2 (Interior evolution). An interior time-evolution is a (strongly continuous) unitary family $U(t)$ on $\mathcal{H}$,

$$U(t)^*U(t) = I, \quad U(t+s) = U(t)U(s), \quad U(0) = I.$$

(If one prefers time-dependent generators, read $U(t)$ as the propagator.)

Definition 3 (Readout space and readout map). An observation/reporting space is a metric space Y (often $Y = \mathbb{R}_{\geq 0}$). A *readout* is a map

$$\mathcal{R} : \mathcal{H} \rightarrow Y.$$

Definition 4 (Gate (admissibility filter)). A *Gate* is a map $G : \mathcal{H} \rightarrow \mathcal{H} \cup \{\perp\}$, where $\perp$ is an absorbing “invalid” state. The *admissible domain* is $\text{Adm} := \{\psi \in \mathcal{H} : G(\psi) \neq \perp\}$.

Definition 5 (Pipeline). The full observed pipeline is

$$y(t) = \mathcal{R}(G(U(t)\psi_0)) \in Y.$$

I.2. Four axioms (the whole TOE in four lines)

Axiom 1 (Unitary Interior). Interior dynamics is unitary: $\psi(t) = U(t)\psi_0$ with $\|\psi(t)\| = \|\psi_0\|$ for all t .

Axiom 2 (ABS-first Readout (branch safety)). The measurable scalar is *ABS-safe*: there exists a bounded linear functional $M : \mathcal{H} \rightarrow \mathbb{C}$ (or bounded operator $M \in \mathcal{B}(\mathcal{H})$ plus a probe) such that

$$\mathcal{R}(\psi) = |M(\psi)| \quad \text{or} \quad \mathcal{R}(\psi) = \|M\psi\|.$$

Axiom 3 (Gate as Domain Cut). There exists an admissibility Gate G implementing a domain cut: inadmissible states map to $\perp$ and remain invalid.

Axiom 4 (Monotone Invariants). There exist functionals (“invariants” in the broad, engineering sense) whose values are nonincreasing/nondecreasing under the gated pipeline.

Remark 1 (Interpretation). **Interior** is conservative. **Contrast** lives in **readout**. **Irreversibility** enters via **Gate**.

II. Unconditional theorems (prove once, reuse everywhere)

Everything below follows *unconditionally* from the axioms above (plus explicitly stated side assumptions such as boundedness).

II.1. No pump/no compressor inside

Theorem 1 (Norm invariance under the interior). *If $U(t)$ is unitary, then $\|U(t)\psi\| = \|\psi\|$ for all $\psi \in \mathcal{H}$ and all t .*

Proof. $\|U(t)\psi\|^2 = \langle U(t)\psi, U(t)\psi \rangle = \langle \psi, U(t)^*U(t)\psi \rangle = \langle \psi, \psi \rangle = \|\psi\|^2$. $\square$

Theorem 2 (ABS readout is phase-stable). *Let M be bounded linear, and define $A(\psi) := |M(\psi)|$. Then $A(e^{i\theta}\psi) = A(\psi)$ for all $\theta \in \mathbb{R}$ and $\psi \in \mathcal{H}$.*

Proof. $M(e^{i\theta}\psi) = e^{i\theta}M(\psi)$ by linearity, so $|M(e^{i\theta}\psi)| = |M(\psi)|$. $\square$

Remark 2. This is the precise reason to prefer ABS-first scalars: phase becomes ill-conditioned near nodes, magnitude does not.

II.2. Moral semigroup = absorbing domain cut

Definition 6 (Absorbing set / forbidden set). Fix a forbidden set $\text{Bad} \subset \mathcal{H}$. Define a projection-like Gate

$$G(\psi) = \begin{cases} \psi, & \psi \notin \text{Bad}, \\ \perp, & \psi \in \text{Bad}. \end{cases}$$

Definition 7 (Induced gated evolution). Define the gated evolution by $T(t) := G \circ U(t)$ acting as $T(t)\psi := G(U(t)\psi)$.

Proposition 1 (Semigroup property with absorption). *Assume $G(\perp) = \perp$ and $G \circ G = G$. Then $(T(t))_{t \geq 0}$ is a semigroup on $\mathcal{H} \cup \{\perp\}$:*

$$T(t+s) = T(t) \circ T(s), \quad t, s \geq 0,$$

and once $\perp$ is reached it is absorbing.

Proof. For admissible ψ , $T(s)\psi = G(U(s)\psi)$. Then

$$T(t)(T(s)\psi) = G(U(t)G(U(s)\psi)).$$

If $G(U(s)\psi) = \perp$, then $T(t)(T(s)\psi) = G(U(t)\perp) = G(\perp) = \perp$. If $G(U(s)\psi) \in \mathcal{H}$ and $G \circ G = G$, then $G(U(t)G(U(s)\psi)) = G(U(t+s)\psi) = T(t+s)\psi$. $\square$

Remark 3. This is the clean mathematical core of “hard asymmetry”: unitarity inside, one-way absorption at the boundary.

II.3. Zeno-limit enforcement (continuous monitoring as domain constraint)

Definition 8 (Zeno-limited evolution). Let Π be an orthogonal projection on $\mathcal{H}$ (onto the admissible subspace). Define

$$U_Z(t) := \lim_{n \rightarrow \infty} (\Pi U(t/n) \Pi)^n,$$

whenever the strong limit exists.

Proposition 2 (Zeno is a Gate, not an energy source). *If $U(t)$ is unitary and Π is a projection, each factor $\Pi U(t/n) \Pi$ is bounded with operator norm ≤ 1 . Hence $\|(\Pi U(t/n) \Pi)^n\| \leq 1$, and any strong limit $U_Z(t)$ is a contraction (non-expansive).*

Proof. $\|\Pi\| = 1$ and $\|U(t/n)\| = 1$, so $\|\Pi U(t/n) \Pi\| \leq \|\Pi\| \|U(t/n)\| \|\Pi\| = 1$. Powers preserve the bound, and strong limits of uniformly bounded operators remain bounded by the same envelope. $\square$

II.4. Legacy invariant (monotone functional)

Definition 9 (Legacy functional). Let $Q : \mathcal{H} \cup \{\perp\} \rightarrow \mathbb{R}_{\geq 0}$ be an “integrity density” with $Q(\perp) = 0$. Define

$$\mathcal{L}(t) := \int_0^t Q(T(\tau)\psi_0) d\tau.$$

Proposition 3 (Monotonicity and absorption). $\mathcal{L}(t)$ is nondecreasing in t , and if $T(\tau)\psi_0 = \perp$ for some τ , then $\mathcal{L}(t)$ becomes constant for $t \geq \tau$.

Proof. Nonnegativity gives monotonicity. Absorption gives $Q(\perp) = 0$ for all later times, hence the integral stops accumulating. $\square$

II.5. Tail control lemma (explicit, checkable)

Definition 10 (Bank kernel and decay). Assume a bank Gram kernel decomposes as

$$G_{\text{bank}}(\Delta) = \sum_{j=0}^J w_j^2 K_j(\Delta), \quad w_j^2 \geq 0,$$

and for a fixed integer $N \geq 4$ there exist constants $C_{j,N} > 0$ such that

$$|K_j(\Delta)| \leq C_{j,N}(1 + |\Delta|)^{-N} \quad (\Delta \in \mathbb{R}).$$

Lemma 1 (Universal majorant).

$$|G_{\text{bank}}(\Delta)| \leq \left(\sum_{j=0}^J w_j^2 C_{j,N} \right) (1 + |\Delta|)^{-N}.$$

Proof. Triangle inequality and the decay assumption give the bound termwise and after summation. $\square$

Definition 11 (Tail constant).

$$T(J) := \sum_{m \geq 1} (m+1) \sup_{|\Delta| \in [m, m+1]} |G_{\text{bank}}(\Delta)|.$$

Theorem 3 (Explicit tail bound). Let $S_J := \sum_{j=0}^J w_j^2 C_{j,N}$. Then

$$T(J) \leq (\zeta(N-1) - 1) S_J.$$

Proof. By the majorant, for $|\Delta| \in [m, m+1]$ we have $(1 + |\Delta|)^{-N} \leq (m+1)^{-N}$, hence

$$\sup_{|\Delta| \in [m, m+1]} |G_{\text{bank}}(\Delta)| \leq S_J (m+1)^{-N}.$$

Therefore

$$T(J) \leq \sum_{m \geq 1} (m+1) S_J (m+1)^{-N} = S_J \sum_{m \geq 1} (m+1)^{-(N-1)} = S_J \sum_{k \geq 2} k^{-(N-1)}.$$

The last sum is $\zeta(N-1) - 1$. $\square$

Proposition 4 (Checkable stability inequality). If a leakage functional obeys

$$\theta(t; J) \leq \frac{(\log t) T(J)}{G_{\text{bank}}(0)} \quad (G_{\text{bank}}(0) > 0),$$

then

$$\theta(t; J) \leq \frac{(\log t) (\zeta(N-1) - 1)}{G_{\text{bank}}(0)} \left(\sum_{j=0}^J w_j^2 C_{j,N} \right),$$

and $\theta(t; J) < 1$ reduces to the explicit inequality

$$\sum_{j=0}^J w_j^2 C_{j,N} < \frac{G_{\text{bank}}(0)}{(\log t) (\zeta(N-1) - 1)}.$$

III. Instantiations (parts): RH, SAT/ P vs NP , Navier–Stokes

This section shows *how* each hard domain plugs into the same TOE spine. We only state unconditional facts and clearly label conjectures/reductions.

III.1. Riemann Hypothesis (RH) as spectral-node discipline

Definition 12 (Prime-log spectrum (frequencies)). For $\text{Re}(s) > 1$,

$$\log \zeta(s) = \sum_p \sum_{r \geq 1} \frac{1}{r} p^{-rs}.$$

Writing $s = \sigma + it$ gives frequencies $\omega_{p,r} = r \log p$ via $p^{-rs} = p^{-r\sigma} e^{-itr \log p}$.

Definition 13 (ABS-safe “zero radar” proxy). Fix cutoffs P, R and define

$$L_{P,R}(t) := \sum_{p \leq P} \sum_{r=1}^R \frac{\log p}{p^{r/2}} e^{-itr \log p}, \quad A(t) := |L_{P,R}(t)|.$$

This is a readout in the TOE sense: structured interior sum, ABS-safe scalar output.

Remark 4. $A(t)$ is not $\zeta(\frac{1}{2} + it)$, but it is an explicit, branch-safe prime-side observable that can be instrumented, windowed, and stress-tested under tail bounds.

Definition 14 (Hilbert–Pólya criterion (reduction)). A Hilbert–Pólya operator is a self-adjoint operator H whose spectrum encodes the imaginary parts of nontrivial zeros.

Proposition 5 (Why self-adjointness would force the critical line (reduction)). *If one constructs a self-adjoint operator H whose spectral determinant reproduces the completed zeta function (in a standard sense), then nontrivial zeros correspond to real spectral parameters, which forces $\text{Re}(\rho) = \frac{1}{2}$.*

Proof sketch (standard reduction). Self-adjoint operators have real spectrum. In Hilbert–Pólya constructions, zeros correspond to spectral points; real spectrum then places them on the symmetry line after the standard change of variables built into the completed zeta function. The hard part is constructing H with the required determinant identity. $\square$

Conjecture 1 (RH). All nontrivial zeros of $\zeta(s)$ satisfy $\text{Re}(s) = \frac{1}{2}$.

III.2. SAT and P vs NP as “witness readout” vs “full interior search”

Definition 15 (Constraint system and witness). A SAT instance is a Boolean formula Φ . A witness is an assignment σ such that $\sigma \models \Phi$.

Proposition 6 (Verification is a Gate + readout). *Given σ , checking $\sigma \models \Phi$ is a bounded-time readout: it is a projection of the huge interior search space onto a finite certificate.*

Remark 5. This is the TOE lens: **search** lives in X (huge interior), **witness checking** lives in Y (small readout).

Conjecture 2 ($P \neq NP$). There is no polynomial-time algorithm deciding SAT on all inputs.

III.3. Navier–Stokes regularity as “Gate invariance” against blow-up

Definition 16 (Navier–Stokes (informal placement)). Navier–Stokes is a nonlinear evolution PDE with a conserved/controlled energy inequality in weak settings. Regularity asks whether smooth initial data stays smooth globally in time.

Remark 6. In TOE terms: the **interior** is nonlinear evolution; the **Gate** is admissibility (bounded norms / no singularity); the question is whether the Gate is *provably invariant* (never violated).

Conjecture 3 (Navier–Stokes regularity (3D)). Smooth initial data yields globally smooth solutions in 3D (or not). This is open.

IV. Concrete numerics as a sanity layer (energy bookkeeping example)

This is a purely arithmetic normalization example (no physics claim).

Definition 17 (Energy mix normalization). Given nonnegative components E_i , define $E_{\text{tot}} = \sum_i E_i$ and normalized shares $M_i := E_i/E_{\text{tot}}$. Then $\sum_i M_i = 1$ exactly (up to machine precision in floating arithmetic).

component	E_i (J)	M_i
modes	1.855×10^{-19}	$\approx 5.0 \times 10^{-17}$
vacuum	3.313×10^{-19}	$\approx 8.9 \times 10^{-17}$
gravity	1×10^{-45}	$\approx 2.7 \times 10^{-43}$
entropy	6.626×10^{-16}	$\approx 1.79 \times 10^{-13}$
correlation	5.831×10^{-19}	$\approx 1.57 \times 10^{-16}$
consciousness	3.711×10^{-3}	≈ 0.99999999999998
emergent	6.626×10^{-17}	$\approx 1.79 \times 10^{-14}$

Remark 7. This table is only a normalized-weight ledger; it does not assert physical ontology for each label.

V. Code comes last: minimal bridge specification

This section is intentionally short and surgical: it tells you what the code must implement *given* the TOE spine.

V.1. Required interfaces (implementation contract)

A compliant implementation exposes:

1. **Interior state** representation ψ (vector, signal bank, or structured object).
2. **Unitary/Conservative step** $U(\cdot)$ (exact unitary or numerically stable surrogate).
3. **Gate** $G(\cdot)$ returning either state or $\perp$.
4. **Readout** $\mathcal{R}(\cdot)$ returning an ABS-safe scalar.
5. **Invariants** (legacy accumulator, leakage bound calculators).

V.2. `n_singular` as the seed (deterministic parameterization)

If the model is “computed from n_{singular} ”, treat it as a deterministic parameter that fixes:

- (i) bank weights w_j , (ii) window constants $C_{j,N}$, (iii) solver schedule / knobs.

The key engineering rule is: *do not* treat it as a proof oracle; it is a reproducible seed.

V.3. Minimal pseudocode

```
psi = init_state(n_singular)
for t in timeline:
    psi = U_step(psi, t)           # conservative interior
    psi = Gate(psi)                # absorbing if forbidden
    y[t] = abs(Readout(psi))       # ABS-first scalar
    legacy[t] = legacy[t-1] + Q(psi)*dt
    theta[t] = leakage_bound(params, t)
```

VI. What I still need from you to weld to your current repo (one line)

Some of your earlier uploaded files have expired from the workspace snapshot. If you want this document to *auto-reference* exact function names/paths and patch points in your current Python files, re-upload the current versions of:

- `DREAM6_operator_INF.py`
- `RH_MADNESS_5_ROBUSTSCORE.py`

Then I will produce a surgical diff: exact lines to change, and nothing else.